Student Course Management System Report

Student Name:

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Project Title:

Student Course Management System

Repo => **https://github.com/rabea-shaban/Student_Course_Management_System_Individual_assignment**

2. Introduction

In modern educational systems, managing student data and course enrollment efficiently is essential. This project presents a **Student Course Management System** developed using **C++** as a console-based application.

The system aims to provide a simple yet effective way to manage students, their academic information, and the courses they are enrolled in. It demonstrates the use of **Object-Oriented Programming (OOP)** principles and **Standard Template Library (STL)** to build a structured and maintainable solution.

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4. Detailed Project Description

The project is a console-based system that allows users to perform various operations related to student management.

Key Functionalities:

- Add a new student with ID, name, and GPA
- Remove a student from the system
- Search for a student using ID
- Display all students
- Enroll a student in courses
- Display student courses
- Sort students based on GPA

The system ensures that:

- GPA values are validated (0–4)
- Duplicate courses are avoided using `set`

5. Project Beneficiaries

This system can benefit:

- **Students** → learning OOP and system design
- **Educational institutions** → basic student tracking
- **Developers** → understanding STL and data structures
- **Beginners in C++** → practical application of concepts

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6. Detailed Analysis

Problem Statement

Managing student records manually is inefficient and prone to errors. A digital system is required to handle operations efficiently.

Proposed Solution

A C++ application using:

- `vector` → to store students
- `set` → to store courses (no duplicates)

Key Functions:

- `findStudent()` → search student
- `addStudent()` → add new student
- `removeStudent()` → delete student
- `sortStudents()` → sort by GPA

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7. Task Breakdown Structure

Task	Description
Design Classes	Person & Student
Implement Features	Add, Remove, Search
Course System	Add & display courses
Sorting	GPA sorting
Testing	Validate all operations

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8. Time Plan

Phase	*Duration*
Planning	1 day
Development	3 days
Testing	1 day

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9. System Architecture and Design

Classes:

Person Class

- Attribute: name
- Method: getName()

Student Class (inherits from Person)

- Attributes: id, gpa, courses
- Methods:
  - addCourse()
  - showCourses()
  - display()

Relationships:

- **Inheritance:** Student → Person

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10. Testing Scenarios and Results

Test Case	Input	Expected Output
Add Student	Valid data	Student added
Invalid GPA	GPA > 4	Error message
Search Student	Existing ID	Display data
Remove Student	Valid ID	Removed
Enroll Course	Valid ID	Course added

All test cases were successfully passed.

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11. End-User Guide

1. Run the program
2. Choose from menu:
    - Add Student
    - Remove Student
    - Search
    - Display
    - Enroll Course
    - Show Courses
    - Sort
3. Enter required data
4. View results

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12. Conclusion

This project successfully demonstrates how to build a structured system using C++. It applies core programming concepts such as OOP, data structures, and algorithms.

The system is simple but scalable and can be extended into a full application with database support and graphical interface.

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13. References

- C++ Documentation
- STL (Standard Template Library)
- GeeksforGeeks
- StackOverflow

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14. Appendices

- Source Code (GitHub Link)
- Flowchart Diagram
- UML Diagram
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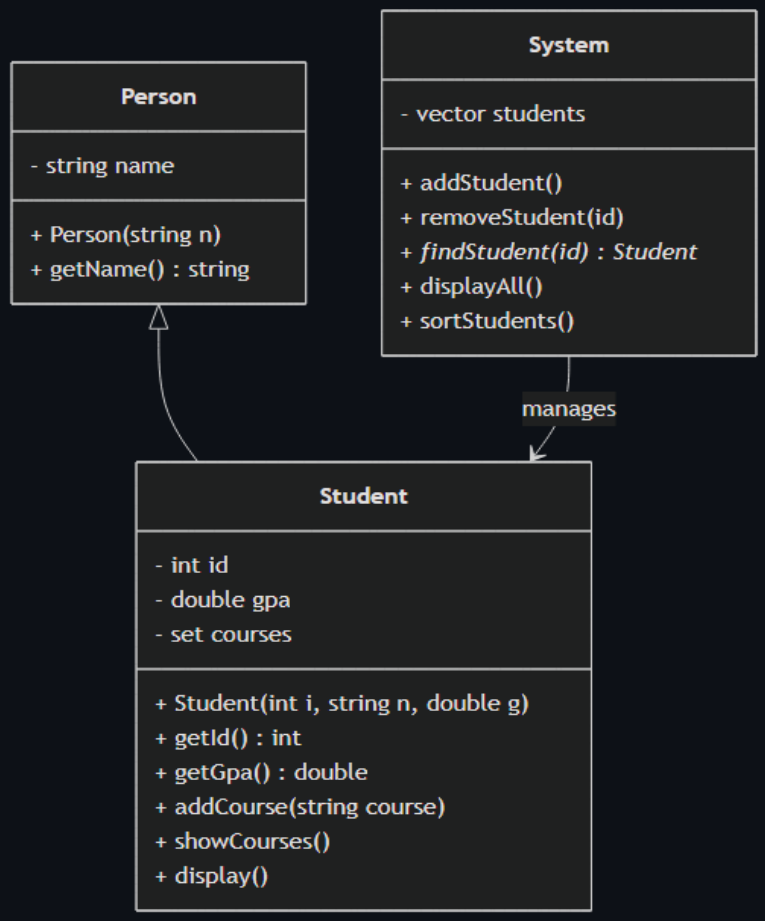
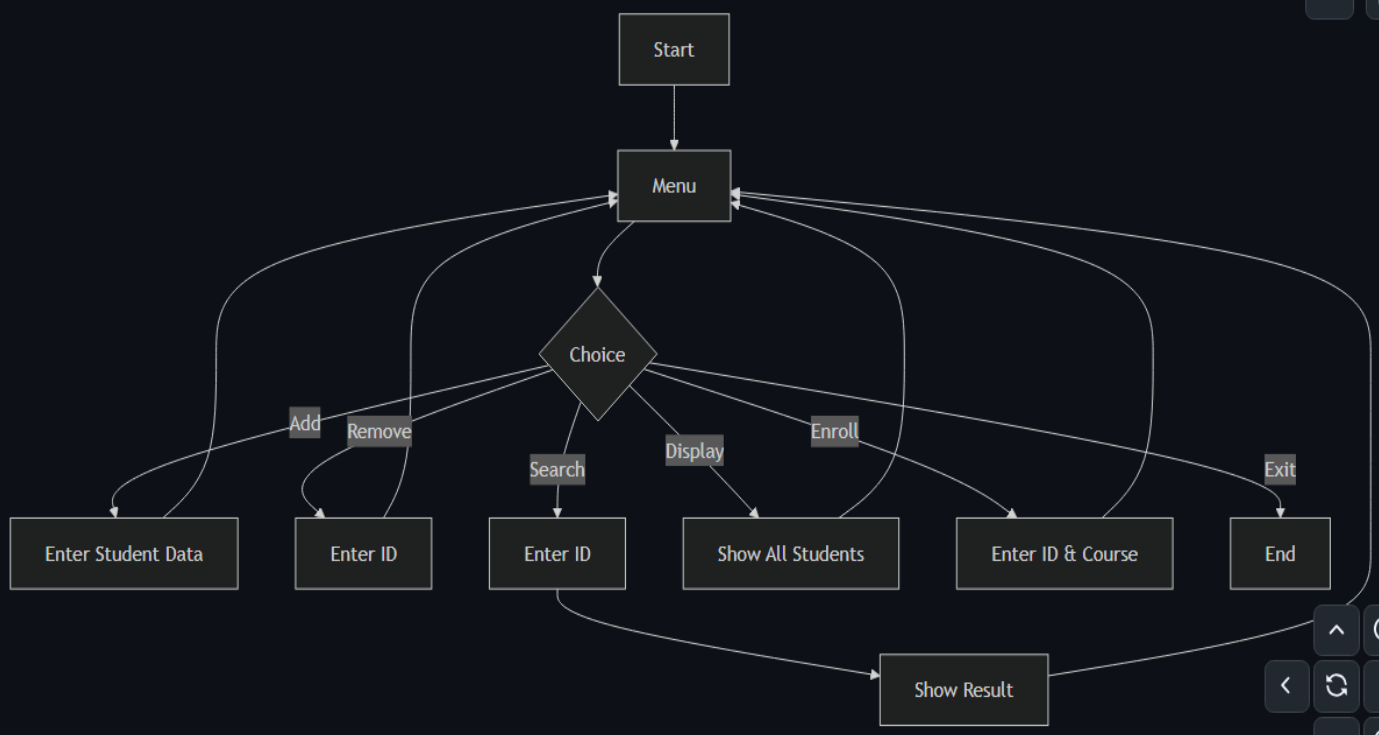
UML Class Diagram



Flowchart